

Assistive and Rehabilitation Robotics

YU Haoyong, PhD (MIT)

Associate Professor

Department of Biomedical Engineering



Biorobotics Lab Overview

Funded by National Robotics Program



Assisting Workers

Construction, logistics,
factories, and farming,
caregivers



Assisting Patients

Stroke Rehabilitation
Daily living assistance

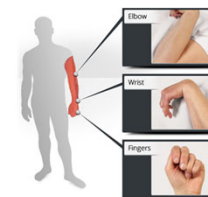
*We develop novel robotic systems, devices, and enabling technologies to address the healthcare and manpower challenges in societies with increasing **aging population**.*

Wearable Robotics Have Potential Applications in Multiple Markets

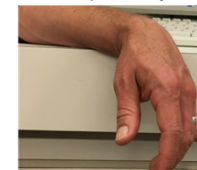


Robots for Stroke Rehabilitation

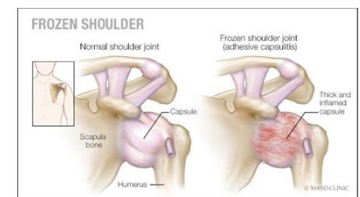
- Stroke increasing rapidly in aging societies
 - 750,000 strokes each year in the US
 - 3 million stroke each year in the china
 - 10,000 per year in Singapore
 - 4th cause of death and 8th leading cause of hospitalization in Singapore
- Recovery through rehabilitation training
 - Principle of neuroplasticity
- Rehab therapy is labor intensive
 - Shortage of therapist



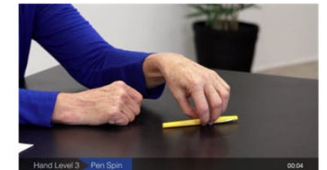
Spasticity



Muscle weakness

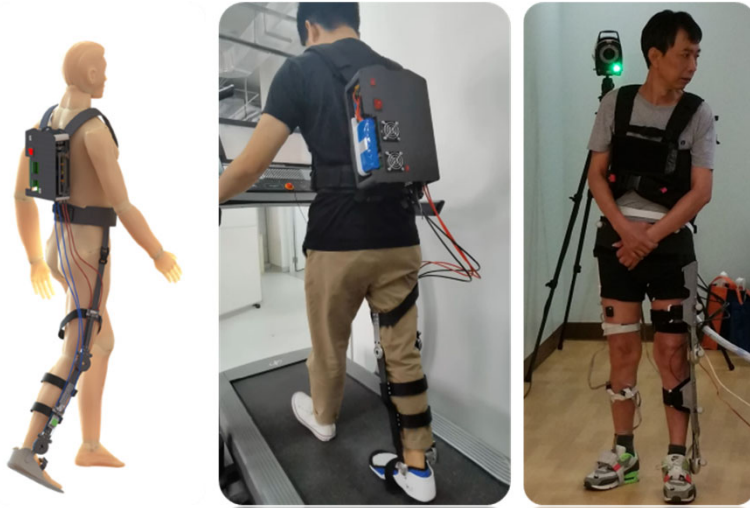


Frozen shoulder



Fine motor skill deficits

A Portable Gait Rehabilitation Exoskeleton



Patented

- **Key advantages**
 - Targeted design for stroke
 - Compliant and safe
 - Patient active control
 - Lightweight and portable
 - Low cost



Current Technologies

Preliminary Clinical Test Result

➤ Subject 1 - videos

Before training



During training



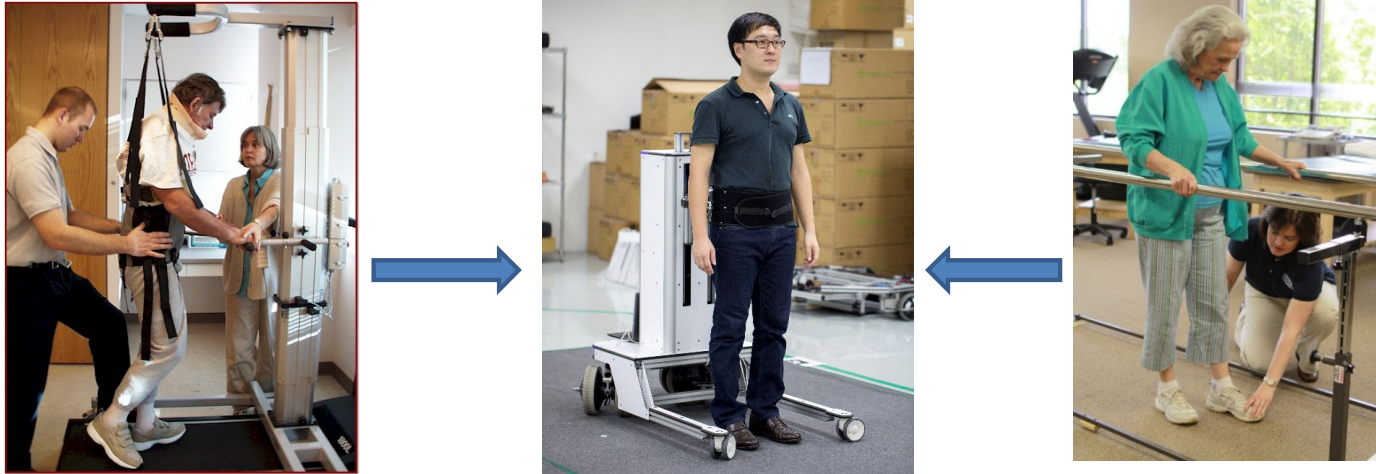
After training for 3 days



Latest Prototype and Clinical Trials



Robotic Walker For Gait Training

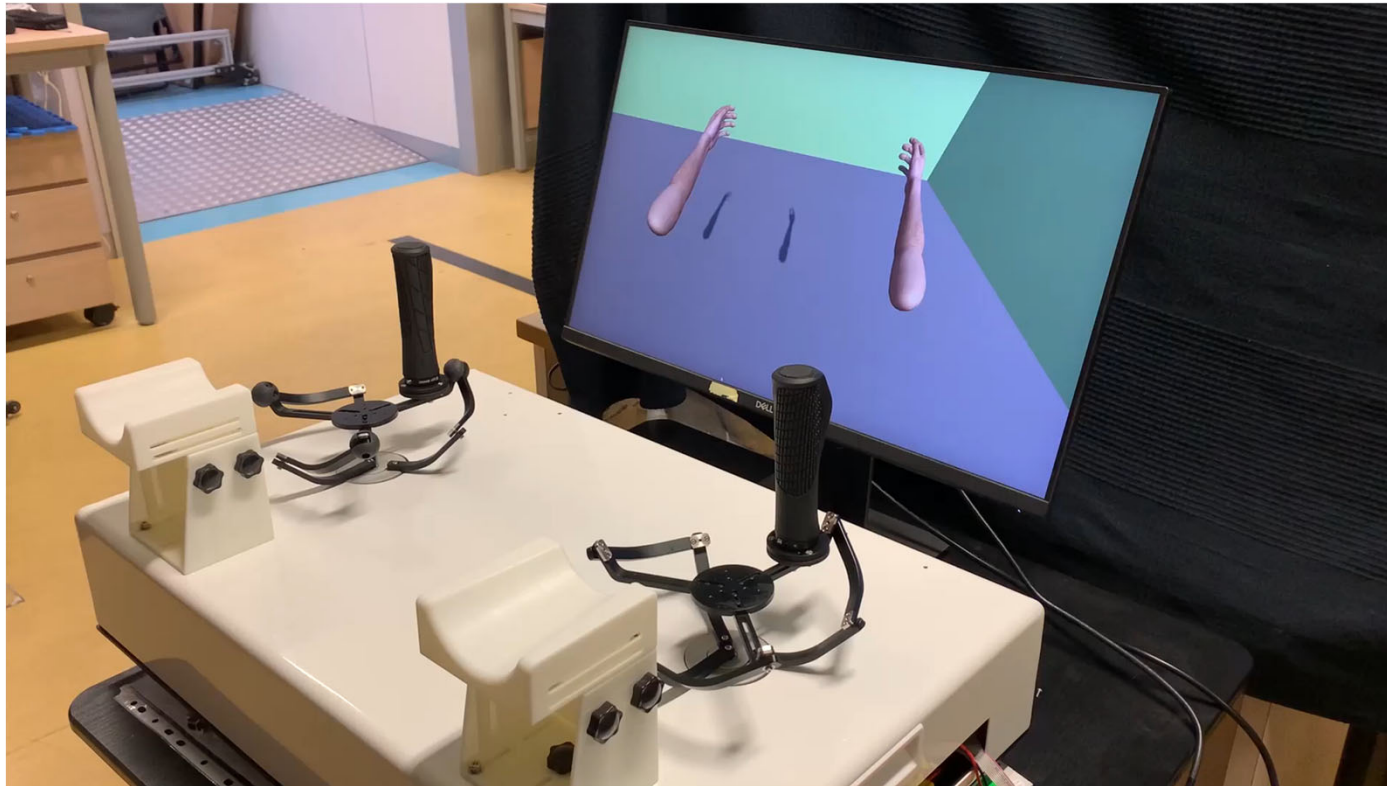


- Relieve therapists from manual labor
- Enable long repetitive training with ease
- Enable more training functions
- Provide better quantitative assessment

Robotic Walker Demonstration



PASSIVE WRIST ROBOT – PROTOTYPE DEMON

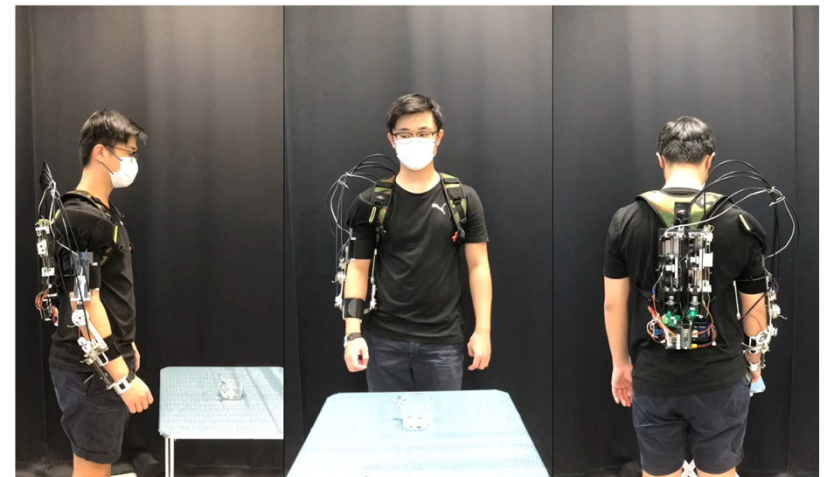


Latest Prototype and Clinical Trials



A Portable Elbow Exoskeleton ADL Training and Assistance

- System concept
 - Two SEAs on the backpack (2Kg)
 - Cable drive to assist elbow movement
 - Assisting ADL (stroke or other muscular functional loss)
 - Distal Arm Training
- Preliminary work
 - Basic hardware prototype tested with basic control
- Proposed Research focus
 - Optimized system design
 - Sensor integration and algorithm development
 - Higher level control



Latest Prototype and Clinical Trials



Elbow Assistive Robot



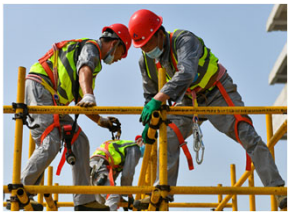
Upper Limb Exoskeleton for Rehab



Increasing Demands for Exoskeleton Technologies

Industrial Protection Scenarios

Exoskeletons reduce risks of musculoskeletal injuries



Construction, logistics, interior renovation, etc. still involve large numbers of physical workers



Global status of musculoskeletal occupational injuries



Occupational injury risks remain high



Aging workforce leads to labor shortage



Exoskeletons can reduce injury risks, extend older workers' employment years, and ease labor shortages [1]

Elderly Assistance Scenarios

Exoskeletons help the elderly regain mobility



canes or wheelchairs is limited in multi-terrain environments; exoskeletons offer stronger support than wheelchairs and canes [2,3].

Outdoor Activities Scenarios

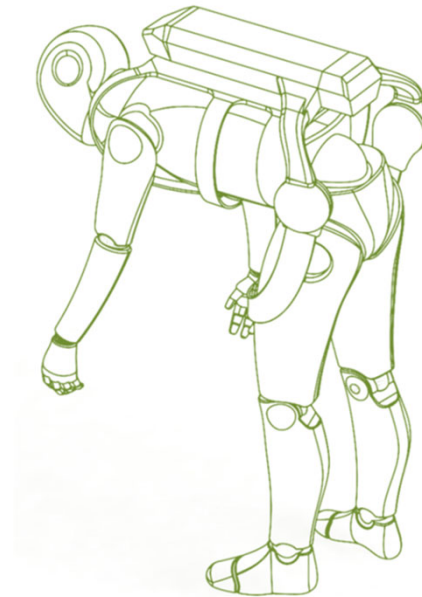
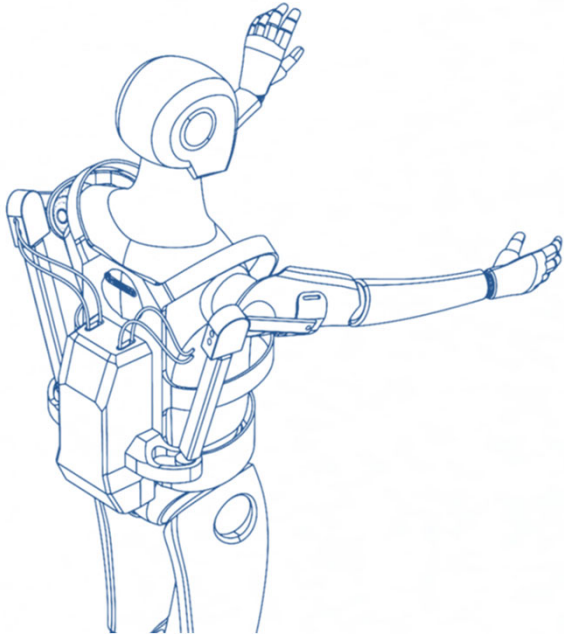
Exoskeletons assist hiking and mountain climbing



Traditional solutions either require brute force endurance or rely on porters to carry people

- [1] A. Baldassarre et al., "Industrial exoskeletons from bench to field: Human-mach and tasks," *Front. Public Health*, vol. 10, Nov. 2022, doi: 10.3389/fpubh.2022.1039680.
[2] C. Siviyy et al., "Opportunities and challenges in the development of exoskeletons for locomotor assistance," *Nat. Biomed. Eng.*, vol. 7, no. 4, pp. 456–472, Apr. 2023, doi: 10.1038/s41551-022-00984-1.
[3] S. Yeem, J. Heo, H. Kim, and Y. Kwon, "Technical Analysis of Exoskeleton Robot," *World Journal of Engineering and Technology*, vol. 7, no. 1, Art. no. 1, Dec. 2018, doi: 10.4236/wjet.2019.71004.

EXOSKELETONS FOR INDUSTRIAL APPLICATIONS





Occupational Exoskeletons for Shoulder and Back Support

Back Support Exoskeletons

Back injury:

- \$13.1 billion costs for injuries related to work in the US(2016)
- Back injury ranks highest, 17.3%



Warehouse



Healthcare

Our Solutions:

- A back support exoskeleton provides up to 50% assistance level for lifting of 25Kg
- Lightweight and easy don/doff
- Unrestrictive to complex human movements



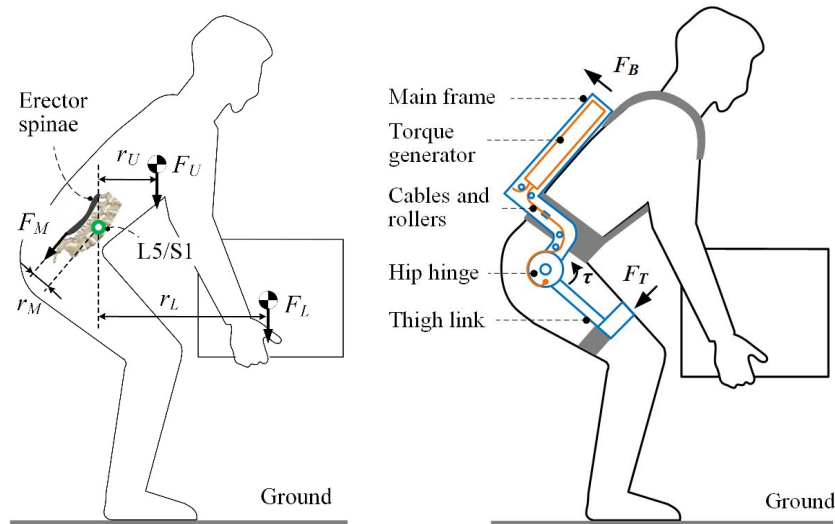
Our Back Support Exoskeleton (Concept, active, passive)

Patented

Back Support Exoskeleton Design

Overall structure

- **Function:** Providing assist torque between trunk link and two thigh links
- **Degree of freedom:** Full waist motion range, hip flexion/extension and abduction/adduction
- **Body fit:** Height and width adjustment, from 1.6m to 1.85m height



Shoulder Support Exoskeletons

Shoulder Injury:

- 13% of all U.S. work-related injuries, second to back injuries



Overhead works with heavy tools



Injury and fatigue of shoulder

Our Solutions:

- Shoulder exoskeleton provides up to 75% assistance level for hand tools up to 5Kg
- Light-weight and easy don/doff
- Unrestrictive to complex human shoulder movements



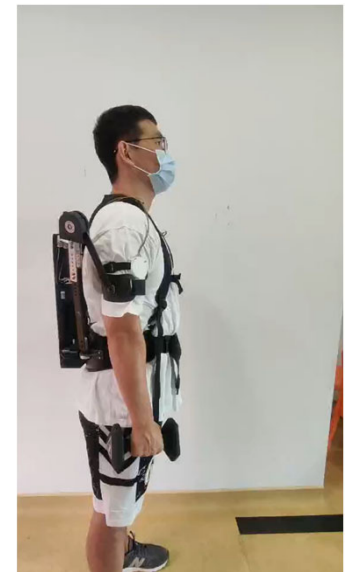
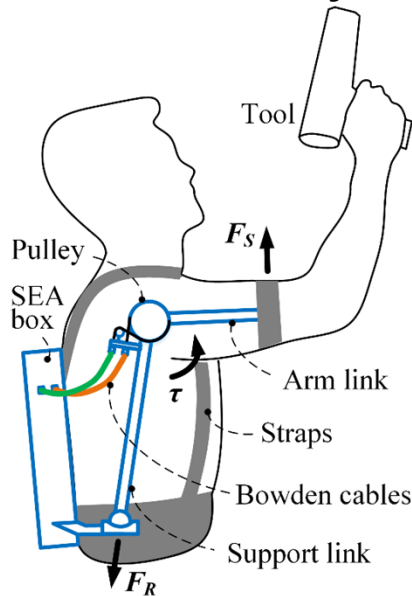
Our Shoulder Support Exoskeleton (Concept, active, passive)

Patented

Active Shoulder Support Exoskeleton

Overall structure

- **Function:** Two SEAs provide assist torque between support link and arm link
- **Degree of freedom:** Cable drive provides full shoulder motion range
- **Body fit:** Height and width adjustment, from 1.6m to 1.85m height



NUS Exoskeletons for Industrial Applications



NUS EXOSKELETONS FOR INDUSTRIAL APPLICATIONS



Latest Prototypes



Potential Markets

Agriculture



Retrieved from: <https://asia.nikkei.com/Politics/Tokyo-wants-experienced-farmers-to-come>

Construction



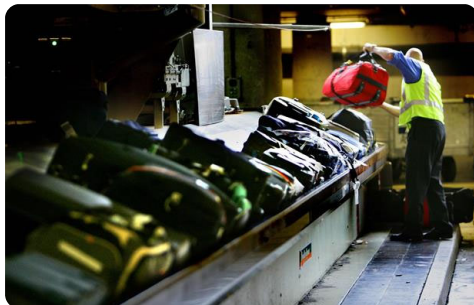
Retrieved from: <https://makinex.com.au/facts-back-injuries-while-working-in-the-construction-industry/>

Warehouse



Retrieved from: <https://www.alberhaskylawfirm.com/>

Aeronautics



Retrieved from: <https://www.exyuaviation.com/2019/09/belgrade-airport-introduces-lost.html>

Automotive



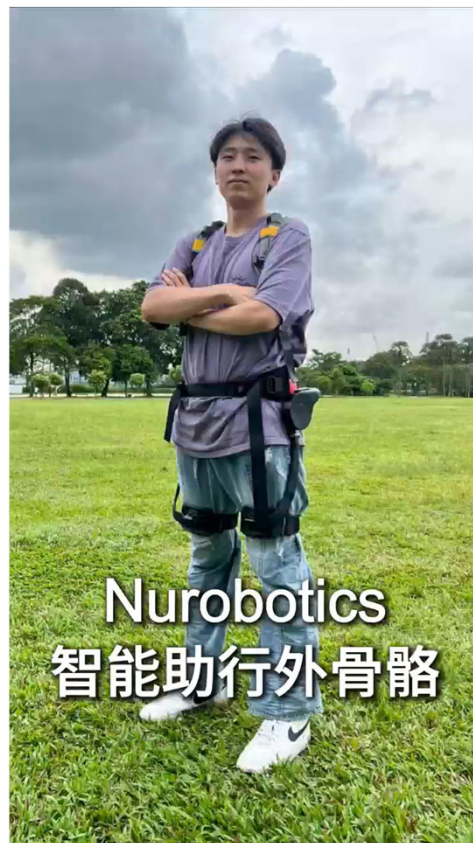
Retrieved from: <https://smallbiztrends.com/2020/09/tire-repair-franchise.html>

Maritime

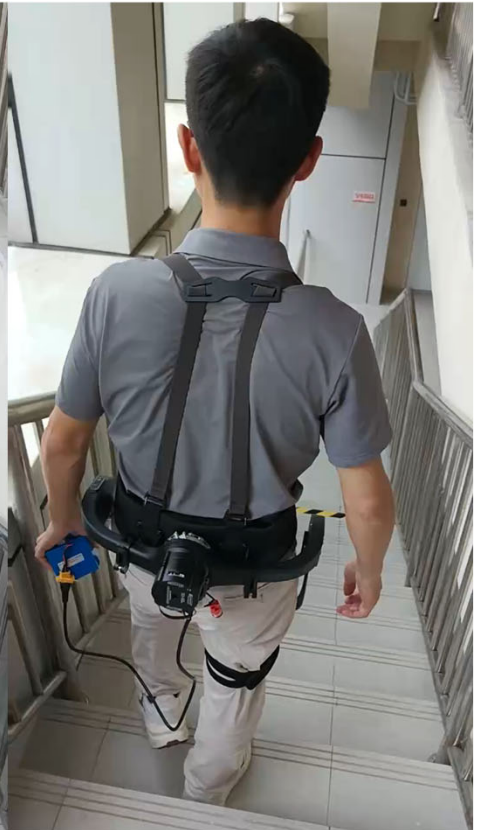


Retrieved from: <https://csupply.se/store/?pdf-catalog=3654>

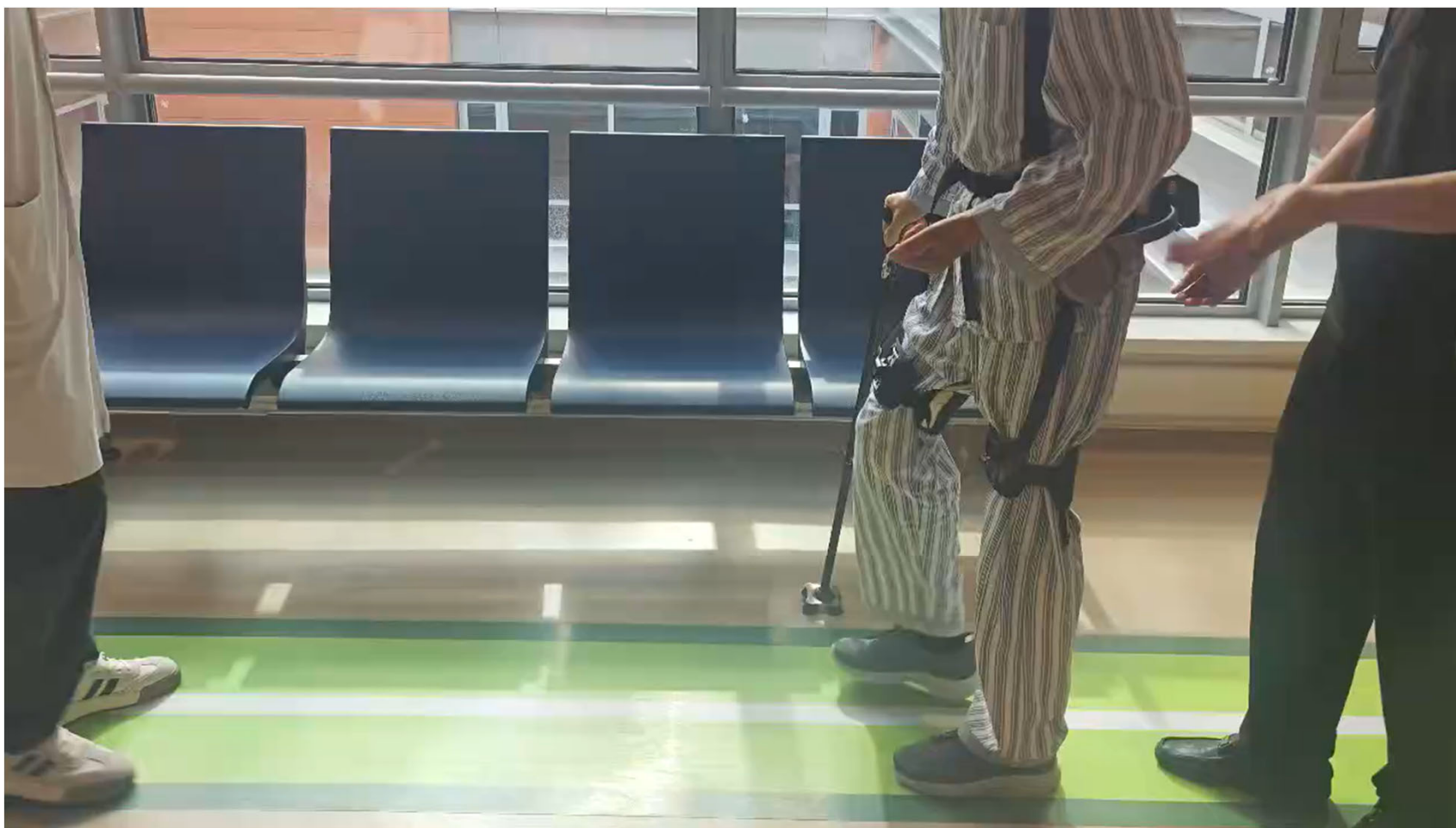
A Terrain and Task-Adaptive Hip Exoskeleton



Hip Exo



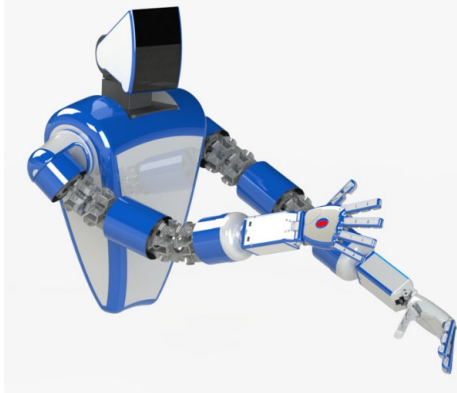
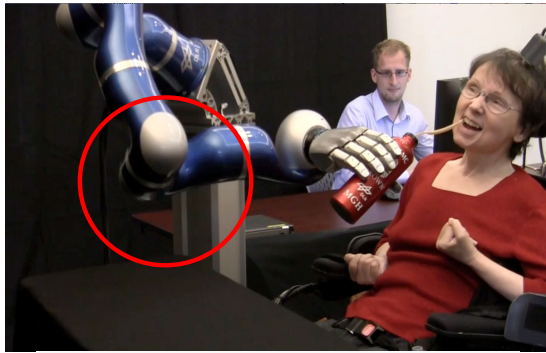
Testing with Stroke Patients



Service Robotics for Healthcare

- **Reduce manpower**
- **Improve productivity**
- **Provide better service**
- **Reduce cost**

Service Robot

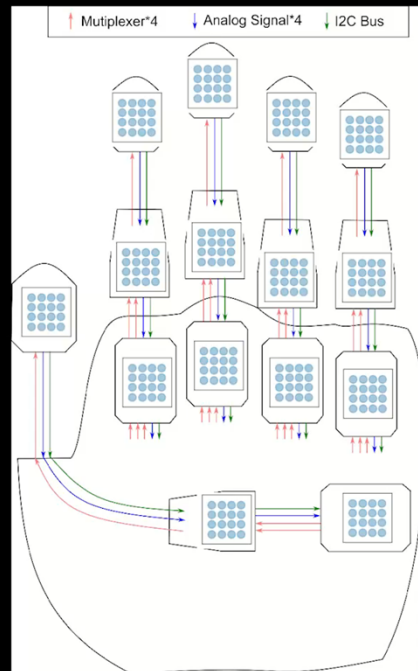


Pouring into a cup

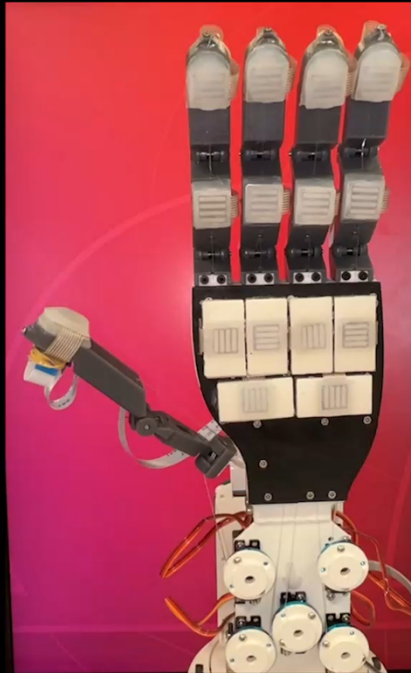
Gtac and Hand Integration

- A robotic hand integrated with GTac at fingers and palm

Real-time tactile feedback



Contacts on the fingers and palm



GTac-Hand: A Robotic Hand with Integrated Biomimetic Tactile Sensing and ECS Recognition Capabilities

Zeyu Lu, Haotian Guo, David Carmona, Shounak Bhattacharya,
and Haoyong Yu

https://roobooot.github.io/GTac_/

Biorobotics Lab

National University of Singapore



Reconfigurable gripper for complex objects grasping

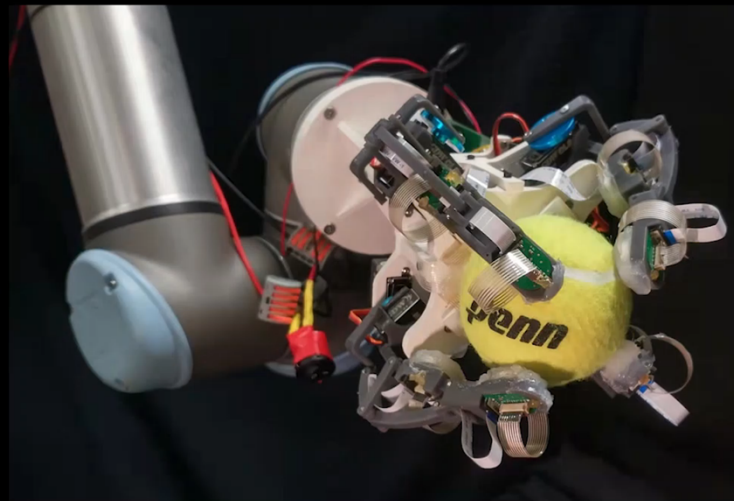
GTac-Gripper:

A Four-Fingered Underactuated Gripper with Reconfigurable Mechanism and Tactile Sensing Capabilities

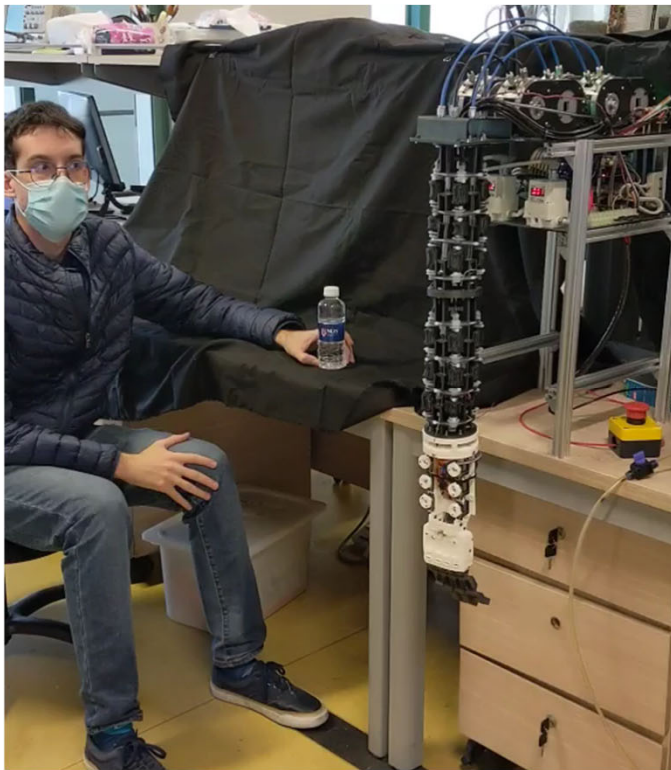
Biorobotics Lab
Faculty of Engineering
Supervisor: Prof Yu Haoyong



NUS
National University
of Singapore

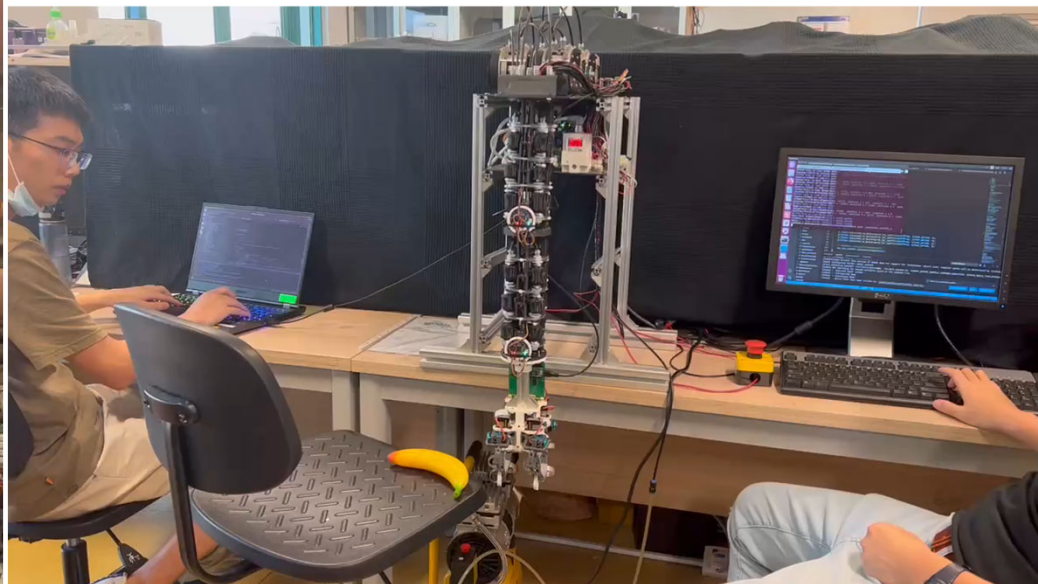
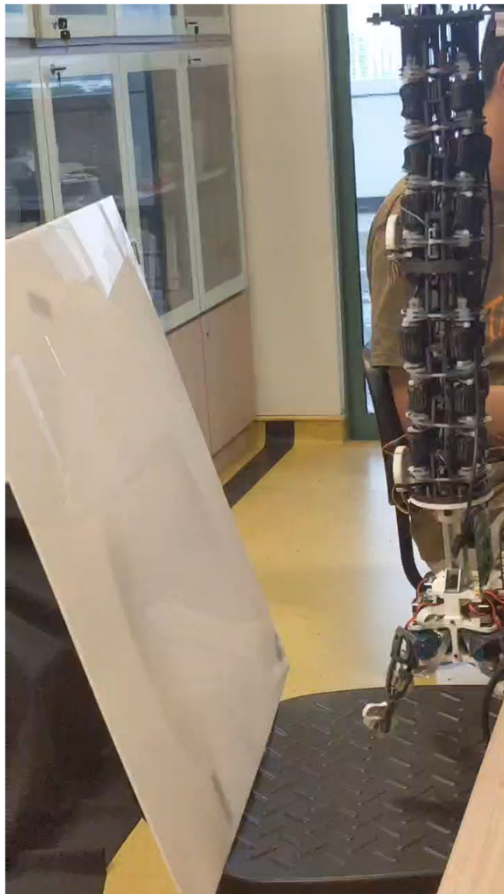


Hybrid Flexible Manipulator



Hybrid Flexible Manipulator

Dynamic modeling and interaction control



**Dynamic interaction with objects
and environment**

Thank You !
